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Spatially Resolved Current Density and Electrochemical Impedance Spectroscopy in PEM Fuel Cells

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ABSTRACT

The durability and efficiency of proton exchange membrane fuel cells (PEMFC) are crucial for their deployment in heavy-duty applications. The membrane-electrode assembly (MEA) of a PEMFC plays a central role and can be characterized using in situ methods like polarization curves, current density mapping, and electrochemical impedance spectroscopy (EIS). In medium-sized to large cells, inhomogeneous conditions often occur across the active area. Segmented current-density measurement boards enable spatially resolved analysis of current density and temperature, and with additional electronics, also spatially resolved EIS. This allows detailed investigation of membrane resistance, diffusion, charge transfer, and proton transport across the cell surface. In this study, current-density distribution under various operating conditions and spatially resolved EIS measurements were performed on an automotive-sized PEMFC. The results show a significant influence of the membrane humidification state on the local performance. Higher humidity, pressure, and stoichiometry improve the homogeneity of the current density distribution. The spatially resolved EIS reveals increasing impedance values from the cathode inlet to the outlet. Equivalent circuit fitting enabled assigning resistances to physical processes and mapping their distributions across the active area. These findings are crucial for a deeper understanding of local operating phenomena in automotive-sized fuel cells.

1 | Introduction

Decarbonizing the transportation sector is crucial to achieving climate targets. Road transport accounts for around 70% of greenhouse gas emissions in the transportation sector. The share of heavy-duty vehicles (HDVs) in greenhouse gas emissions from road transport is around 25% [1, 2]. Polymer electrolyte membrane fuel cells are a promising alternative to combustion engines for reducing greenhouse gas emissions and offer advantages over batteries, especially in the medium and long-distance range, due to their greater range and faster charging times [3, 4]. The long-term behavior and reliability of PEMFCs play a key role in their long-term use. The MEA is a central component of PEMFCs and provides essential insights into mechanisms relevant to efficiency and degradation. Even at the beginning of life (BoL), medium to large fuel cells exhibit heterogeneities across the entire active area, which have a significant impact on performance. These inhomogeneities arise, inter alia, from an uneven distribution of reactant concentrations along the active area, which results from flow patterns, pressure losses, and locally varying reaction rates and has already been described in detail in the literature [5–8]. Furthermore, the formation of local heterogeneities is significantly influenced by an inhomogeneous distribution of water and moisture within the MEA, as described in the literature, for example, by Mitzel et al. [5] and Hussaini et al. [9]. A current density distribution measurement board (CDDM) is suitable for characterizing these inhomogeneities, as it enables spatially resolved recording of the current density and temper-

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ature distribution during the measurement of in situ methods such as polarization curves and sensitivity analyses. Meyer et al. [10] were the first to report on the combined measurement of current density and temperature distribution. In their study, they investigated the mutual influence of temperature and current density in an air-cooled fuel cell operating in air/hydrogen cross-flow mode. Many studies have attempted to improve the homogeneity of current density distribution by making specific adjustments to the material parameters, for example, by varying the porosity of the gas diffusion layer (GDL) [11, 12] or by changing the platinum content of the MEA [11, 13, 14]. In addition, various studies have shown that modifying the flow field design can achieve a more uniform current density distribution [15, 16]. In addition to these design approaches, the influence of operating conditions is also an important research focus, as operating conditions such as temperature, humidity, or gas composition can significantly influence the local current density. Mitzel et al. [5] investigated the influence of stoichiometry and moisture content of the cathode air on the current density distribution in a 10-cell short stack with an active area of 96 cm^2. They showed that even small changes in humidity can have a major impact on the performance homogeneity of the stack and that critical operating conditions, such as local drying out of cells, can be detected via the current density distribution. Zhang et al. [6] also investigated the influence of stoichiometry and humidity on the homogeneity of current density distribution, and additionally examined the influence of back pressure. Their results showed that the current density distribution becomes more homogeneous up to a humidity of 70%, as well as with increasing stoichiometry and increasing back pressure. However, they also found that the humidification rate increases with increasing back pressure, which in turn promotes increased mass transport losses due to flooding. Further investigations into the influence of stoichiometry on current density distribution were conducted by Yin et al. [17] and Yin et al. [18]. Tang et al. [19], on the other hand, investigated the influence of temperature in the range from 50°C to 90°C and identified 70°C as the optimal operating point for both performance and homogeneity of current density. Finally, Liu et al. [20] emphasized that water distribution—and thus moisture balance—is the most sensitive parameter for the performance and stability of fuel cells.

EIS is a key analytical method for characterizing PEMFCs, alongside stationary in situ methods such as polarization curves. By frequency-resolved separation of superimposed processes, EIS allows the identification and quantification of transport, conduction, and reaction processes that occur on different time scales. This makes it possible to specifically assign and distinguish individual loss mechanisms from one another. Numerous studies have been published on the use of EIS measurements to investigate degradation mechanisms [21, 22] or the influence of operating conditions such as temperature or stoichiometry [5, 6, 23–25]. Impedance measurements at the cell level provide valuable information on global transport and reaction mechanisms. However, the design of the bipolar plates and the resulting flow distribution and water balance influence the local performance of the fuel cell, which is not captured by conventional cell EIS measurements. It is therefore very important to determine the impedance spatially resolved over the entire active area. Numerous studies, including work by Schneider et al. [26–28], Gerteisen et al. [29, 30], and Liu et al. [31, 32], have shown that spatially resolved EIS measurements provide crucial insights into local differences and limitations in cell performance. Much of the previous work on spatially resolved EIS has focused on the influence of the flow field on the impedance distribution. These studies investigate, for example, how different flow fields influence the gas distribution and thus the electrochemical reactions [29, 30, 33]. Another key research topic is the water balance within the fuel cell. This involves investigating how gas humidification affects cell performance and the effects of uneven water distribution on electrochemical processes [28, 31, 32]. In most of the published studies on spatially resolved EIS measurements, only small cell formats have been analyzed. Only a few studies have focused on spatially resolved EIS measurements on large-format full cells, which are of greater relevance for practical use in fuel cell systems. One of the few comprehensive studies in this area was carried out by Haimerl et al. [34], who analyzed in detail the influence of operating conditions on the local impedance distribution and ageing effects in a freeze-cycle test on a PEM fuel cell with an active area of 285 cm^2. There are currently two established methods for measuring spatially resolved impedances in fuel cells. The most widely used method for measuring local impedances is based on the use of segmented cells [26–29]. An alternative method is to integrate a CDDM into the measurement setup [31, 32, 34].

This work contributes to the expansion of knowledge about local impedances of automotive-sized PEM fuel cells. In contrast to previous approaches, in which the local impedance was mainly recorded via a segmented cell, a CDDM is used here. Segmented cells make it possible to divide the active area into electrically separate areas, allowing local currents and voltages to be recorded separately. However, segmentation can influence cell behavior, particularly regarding gas flows, water management, and heat transfer. The use of a CDDM ensures that these influences are minimized: Gas distribution and thermal properties remain largely unchanged compared with normal stack operation, which enables a realistic investigation of cell behavior [31]. However, it should be noted that with CDDM, the individual measurement zones are not completely electrically decoupled from each other, which means that slight interactions between neighboring areas can occur.

Special extension boards for the CDDM allow the local impedance to be recorded directly on the measurement segments. The measuring segments can be read out individually or combined into larger measuring ranges using solder connections.

2 | Experimental

2.1 | Test-Setup

A 10-cell short stack with an active area of 280 cm^2 was used for all investigations. The "generic" open-design metal stack hardware [35] used was originally developed as part of the HyFaB project and is now being used in the RealHyFC project as an open-design metal stack (ODM). The stack is equipped with a project-defined MEA from the RealHyFC project. The tests were performed using a counter-flow method, in which the coolant flows in the same direction as the cathode gas. Deionized water was used as a

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coolant. The tests were performed on a Greenlight (Burnaby, BC, Canada) G200 test bench in galvanostatic mode.

A CDDM from S++ (Murnau-Westried, Germany) was integrated between cells 5 and 6. The measurement board has 667 measurement segments arranged in a 29 × 23 matrix for high-resolution current density mapping. The measuring area of the CDDM is slightly larger than the active cell area. For this reason, the boundary segments of the CDDM were excluded from the evaluation, and the measured currents were taken into account with a correction factor in the other cells. The calculation of the correction factor is shown in Equation (1).

$$correction\,\text{factor} = \frac{j_{wo\,\text{boundary cells}}}{j_{\text{total}}} \tag{1}$$

The correction factor is calculated from the sum of the current density of all measurement segments, without the boundary segments $j_{wo\ \text{boundary cells}}$ in relation to the total current density of all measurement segments j_{total}. The corrected current density in the respective measurement segment $j_{\text{segment, corrected}}$ is calculated from the measured current density in the respective segment $j_{segment,\ measured}$ divided by the correction factor (see Equation 2).

$$j_{\text{segment, corrected}} = \frac{j_{\text{segment, measured}}}{\text{correction factor}} \tag{2}$$

Before electrochemical characterization, the stack underwent a break-in procedure based on the protocol from PowerCell Sweden AB used in the RealHyFC project. This serves to condition the stack to achieve the full performance of the fuel cells. Initially, the stack was heated to 68°C under fully humidified nitrogen. The gas media were then changed to 100% hydrogen at the anode and air at the cathode, also fully humidified. Within 30 min, the load was gradually increased to 420 A (1.5 A·cm^{-2}), followed by a holding phase of 3 min. After that, 20 cycles were performed, consisting of alternating between hydrogen soaks and operation at 420 A. For the hydrogen soaks, the load is reduced to 0 A with a ramp of 15 A·s^{-1}. The cathode air was then depleted so that the hydrogen soak could be initiated. After all cell voltages had dropped below 0.05 V, this state was maintained for a further 3 min. The load was then increased again to 420 A with a ramp of 7.5 A·s^{-1} under full media flow and held for 6 min. After completing the 20 cycles, the stack was shut down and left in a resting state for at least 4 h.

2.2 | Characterization Methods

The characterization methods were performed following the break-in. First, six different polarization curves defined in the project were completed. This was followed by the sensitivity test and, finally, the electrochemical impedance measurements at the cell level and spatially resolved. Between all characterization sequences, the stack had a minimum rest period of 8 h to rule out reversible cell voltage losses [36].

2.2.1 | Polarization Curves

In the following analysis, only one of the six polarization curves under standard conditions is examined in detail. The nominal operating conditions specified in the RealHyFC project were used as standard conditions (see Table 1). These parameters were kept constant throughout the measurement. The coolant flow was selected so that the difference between the coolant inlet and outlet temperatures was 10 K at a load point of 1.8 A·cm^{-2}. This coolant flow was kept constant for the other load points. The polarization curve comprised a total of 16 load points in the range from 0 to 1.8 A·cm^{-2}. The measurement was first performed in descending order and then in ascending order. All load points above the minimum load point for stoichiometric operation (0.3 A·cm^{-2}) were held for 5 min each, while load points below this limit were held for 2 min each. Before recording each polarization curve, the stack was conditioned for 30 min at 1 A·cm^{-2} under the corresponding operating conditions.

2.2.2 | Sensitivity Test

To investigate the heterogeneities, a sensitivity study was conducted in which the operating parameters temperature, pressure, humidity, and stoichiometry were systematically varied. Only one parameter was changed at a time, while the other parameters were kept at a constant reference value. Table 2 shows the range of operating conditions that were taken into account in the sensitivity test. The reference values are marked in blue and correspond to the nominal ReaHyFC conditions. A total of 13 different operating states were analyzed, with measurements taken at four load points for each of these states.

For all parameter variations, the gas temperatures, gas mixture, and minimum current for stoichiometric operation were selected according to the nominal conditions (cf. Table 1). The coolant flow at 1.8 and 2.3 A·cm^{-2} was selected so that the temperature difference between the coolant inlet and outlet did not exceed 10 K. At the lower two load points, the coolant flow was kept constant in accordance with the coolant flow at 1.8 A·cm^{-2}.

2.2.3 | Electrochemical Impedance Spectroscopy

The EIS measurements at the cell level and spatially resolved are performed using a Zahner (Kronach, Germany) potentiostat of type Zennium X in combination with a Zahner external load of type EL1002. The external load EL1002 handles the variable load component. The testbench load of type EA-ELR 10080-1000 from the EA Elektro-Automatik GmbH & Co. KG (Viersen, Germany) was used for the static load component. By using an additional 1 V-PAD4 card, up to 16 impedance measurements can be performed in parallel in addition to the reference impedance. For the spatially resolved EIS measurements, the active area was divided into 16 regions in which the measurements could be recorded in parallel. This number cannot be exceeded due to the limitations of the measurement device. The division of these areas was chosen based on the current density distribution under nominal conditions at 1.8 A·cm^{-2}. On the one hand, it was ensured that the current density within each zone is as homogeneous as possible. On the other hand, care was taken that the relevant areas showing inhomogeneous behavior over the cell, including the cathode and anode inlets and outlets, are covered. As shown in Figure 1, seven measurement zones in the cathode

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TABLE 1 | RealHyFC Nominal operating conditions for the polarization curve and for the EIS measurements.

	Inlet temperature/°C	Humidity/%	Outlet pressure/kPaG	Inlet pressure/kPaG	Stoichiometry/-	Gas composition
Anode	75	50	160	—	1.5	80% H_2/20% N_2
Cathode	75	35	150	—	1.7	100% air
Coolant	70	—	—	50	—	—

TABLE 2 | Parameter change in the sensitivity test, with the nominal parameters marked in blue. Simultaneous change of anode and cathode (A/C) for pressure, humidity, and stoichiometry.

Parameter/Variation	1	2	3	4	5
Temperature/°C (coolant inlet)	55	60	65	70	75
Pressure A/C/kPaG (outlet)	60/50	110/100	160/150	210/200	
Humidity A/C/%	30/30	50/35	70/60	90/90	
Stoichiometry A/C/—	1.4/1.6	1.5/1.7	1.9/2.1		
Current density/A·cm^{-2}	0.36	0.9	1.8	2.3	

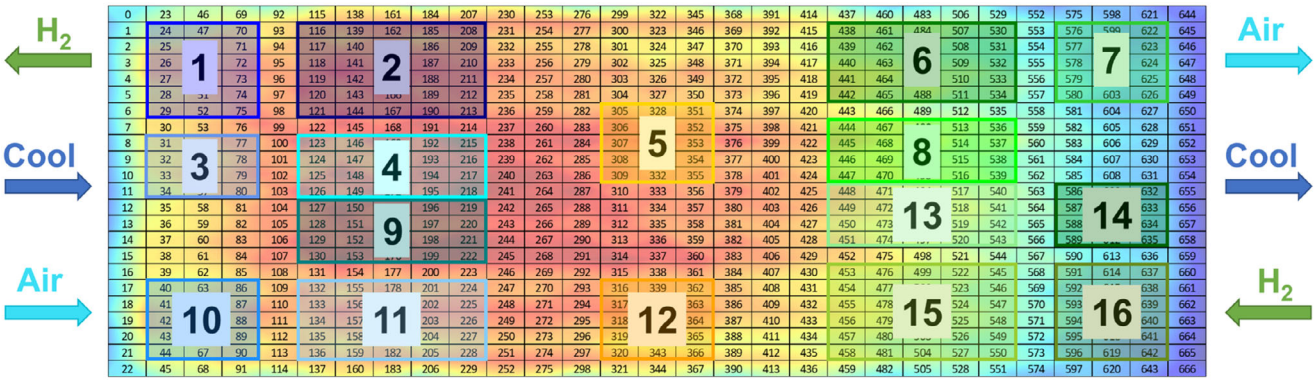


FIGURE 1 | Division of the CDDM measuring cells into a total of 16 different zones across the active area of the fuel cell based on the current density distribution under reference conditions at 1.8 A·cm^{-2} (cf. color code).

inlet and outlet areas and two additional zones in the middle area are taken into account.

The spatially resolved EIS measurements and the single cell measurements are performed galvanostatically. However, the underlying measured variable differs between the two methods. In the single-cell measurements, it is assumed that all cells have the same current density. The composite of the five cells above the CDDM serves as the reference impedance, while the voltage of each cell is recorded as the measured variable. In contrast, the spatially resolved impedance measurements are based on the assumption that the active area acts as an equipotential surface and therefore the same voltage is applied across the entire area. This assumption is justified due to the high electrical conductivity of the metallic bipolar plates. During the spatially resolved measurement, the voltage of cell 6, which is located directly above the CDDM, is recorded as the reference voltage. This voltage is identical for all zones and therefore does not serve as a zone data source, but only as a reference for the galvanostatic stimulation. The zone signals relevant for impedance determination are the locally measured currents. For each of the defined zones, the time-dependent current change caused by the galvanostatic excitation is recorded with the CDDM and transmitted to the Zahner EIS device via special expansion boards for the CDDM. The impedances of the individual zones are determined from the ratio of this local current response to the global reference voltage.

First, impedance measurements were performed on the individual cells at all load points, followed by spatially resolved impedance measurements. All EIS measurements were performed under nominal operating conditions. The stack was conditioned for 30 min before measurement. The conditioning time between load changes was 15 min. The impedance measurements were performed in a frequency range of 5 kHz to 100 mHz. The alternating current (AC) for the single-cell EIS measurements was 5% of the direct current (DC) and 7% of the DC for the spatially resolved measurements. The AC was set slightly higher for the spatially resolved measurements, as this resulted in less noise in the measurement results. The amplitude of the AC in both the cell EIS measurement and the spatially resolved measurement is in the range of 5% to 10% of the DC, and it can therefore be assumed that the linearity requirements are

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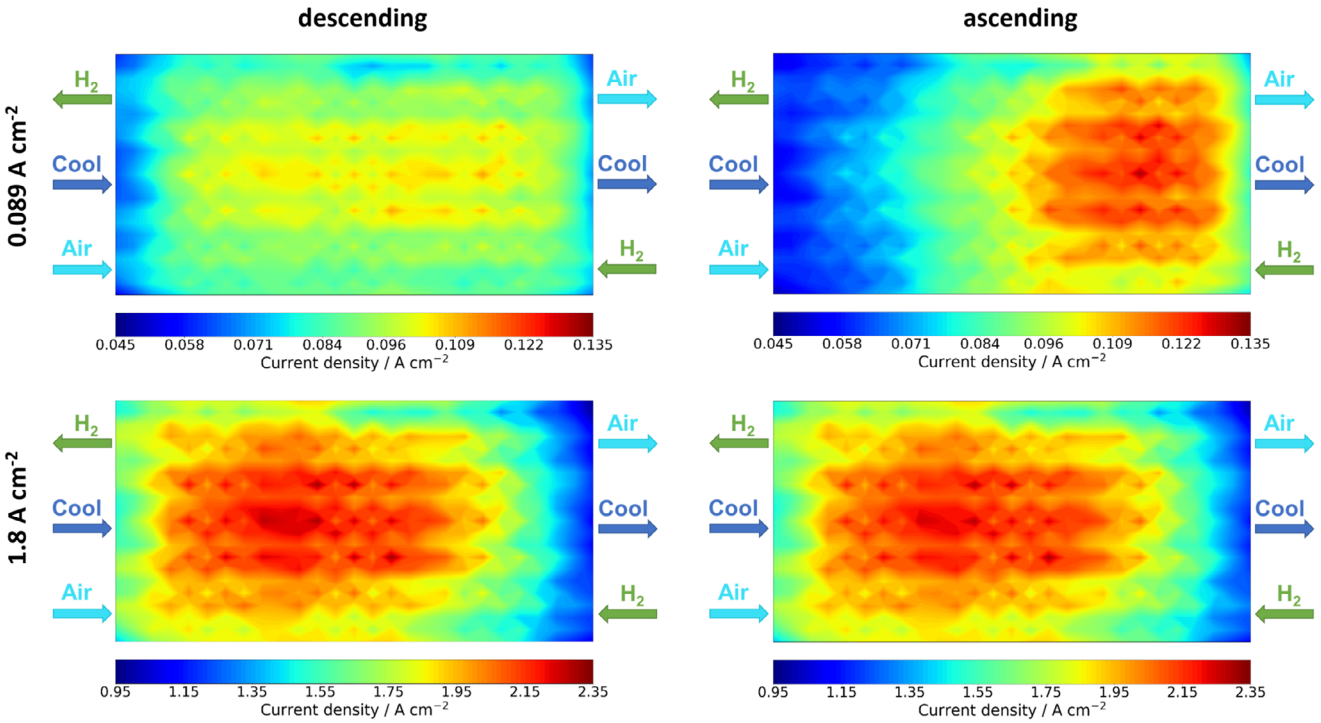


FIGURE 2 | Current density distribution at 0.089 A·cm^{-2} (top) and 1.8 A·cm^{-2} (bottom) for a descending (left) and ascending (right) polarization curve under nominal conditions.

met [37]. The minimum gas flow was set to the respective DC. At the beginning of life (BoL), impedance measurements were performed at 0.18, 0.45, 0.9, 1.44, and 1.8 A·cm^{-2}.

3 | Results and Discussion

The results of the spatially resolved current density distribution and impedance measurements at the cell level and spatially resolved are presented below.

3.1 | Current Density Distribution

Figure 2 shows the current density distributions at load points of 0.089 and 1.8 A·cm^{-2} for the descending and ascending branches of the nominal polarization curve.

Figure 2 helps to understand the different behaviors and the resulting cell voltage for descending and ascending branches of the polarization curve. Comparison of the current density distribution at low and high load points shows a significant influence of the different local conditions, in particular the wetting state of the membrane. At the low-load point shown, a clear difference in the current density distribution can be seen, with a shift of the current density maximum toward the cathode outlet in the ascending polarization curve. This results from over-stoichiometric operation at low-load points, which leads to partial drying out of the membrane. All load points below the minimum load for stoichiometric operation of 0.3 A·cm^{-2} were therefore operated in over-stoichiometric mode. As the membrane dries out, the membrane resistance increases due to reduced proton conductivity, and the current density consequently decreases [5, 6, 9]. Along the descending branch of the polarization curve, a higher amount of residual water remains within the MEA from the preceding high-load operating point, where more product water was produced. This additional hydration is not immediately removed when the load is reduced, resulting in a more uniformly wetted membrane. On the ascending branch, however, the cell approaches the same load point from a lower load operating condition, possibly including over-stoichiometric flows, leading to lower membrane hydration and consequently higher local resistance. This effect has already been described several times in the literature [38, 39]. At high load points, there is no difference in the current density distribution of the two branches. Here, the membrane is moistened by the product water to the same extent in the ascending polarization curve as in the descending polarization curve.

Locally resolved polarization curves can be used for further analysis of the gradients within the current density distribution. These polarization curves are calculated by averaging the current values from the individual measurement segments within a defined zone and plotted against the average stack voltage. Figure 3 shows the descending branch of the nominal polarization curve represented as spatially resolved polarization curves of the 16 zones defined in Figure 1.

Figure 3 illustrates the heterogeneity of local performance across the active area. With a constant value for the average cell voltage of 0.618 V (corresponding to the highest load point of 1.8 A·cm^{-2}), the maximum difference between the inlet areas (marked by blue-based colors) with a deviation of 0.442 A·cm^{-2} is smaller than between the outlet areas (marked by green-based colors) with 0.761 A·cm^{-2}. The largest difference in current density between the zones is found between zones 4 and 7. At a load point of

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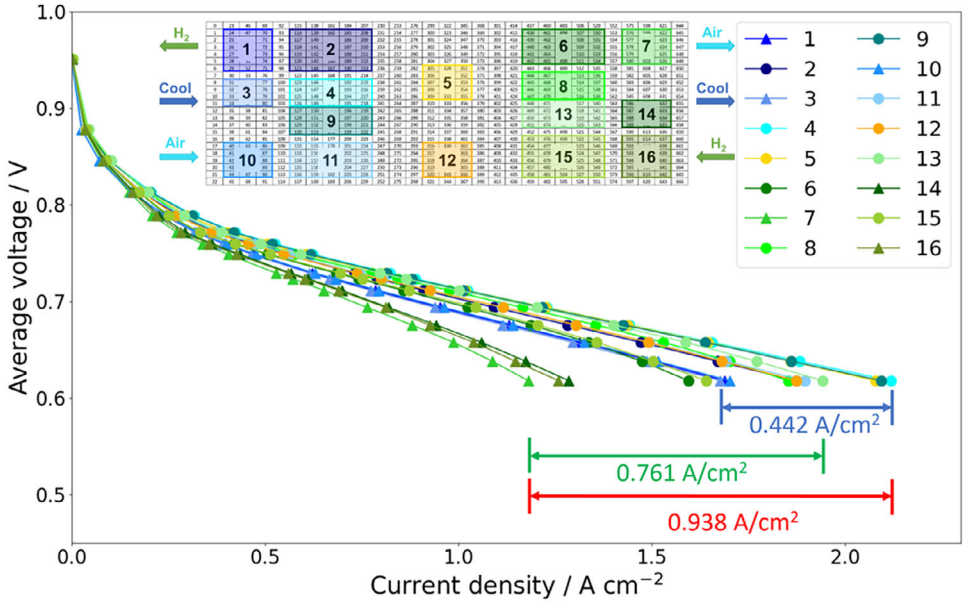


FIGURE 3 | Spatially resolved current density distribution over the 16 different zones of the active cell surface (cf. Figure 1) during a polarization curve under nominal conditions, plotted as a U-I curve where the local current density values at a fixed voltage value, respectively, were averaged over the CDDM measuring cells of each zone.

1.8 A·cm^{-2}, this deviation is 0.938 A·cm^{-2}. The maximum spread in the current density distribution of more than 0.9 A·cm^{-2} is quite common for a stack with an active area of 280 cm^2. Reports in the literature describe stacks of a similar size that exhibit comparable spreads in current density distribution [34, 40, 41], though usually the respective stack design employed is not disclosed. Overall, the generic HyFaB stack can be considered state-of-the-art compared with present commercial stack designs. As shown in the following sections, a more homogeneous current density distribution can be achieved by adjusting the operating conditions. Further improvements may be achieved by modifying the design of the bipolar plates. An important prerequisite for uniform current density distribution is a homogeneous oxygen partial pressure across the entire active area. Zhang et al. [16] have shown in their work that a baffled flow field can make the current density distribution more uniform. Significant improvements in the uniformity of the oxygen partial pressure may furthermore be achieved by using so-called microflow fields, which employ comparatively narrow fins. The narrow fins reduce the restrictions on mass transport and allow for improving the local oxygen concentration, which is especially relevant at the cathode outlet [42]. In the outlet zones, it is clearly visible that the characteristic curves begin to deviate from the linear form at high current densities. This indicates the local occurrence of diffusion losses, which point to a limited mass transport capacity in these areas [25, 43]. In contrast, the middle zones and the inlet zones still show predominantly linear behavior even at high current densities, which suggests that ohmic losses dominate there and that mass transport is not yet limiting.

The maximum current density deviations between the defined zones were further analyzed as part of the sensitivity study. The largest deviation occurred between zones 4 and 7, which is why these zones were selected in the following. The corresponding values are shown in Figure 4. In the sensitivity analysis, the current density was only increased as long as the voltage in no cell of the stack fell below 500 mV. In addition, the highest load point in combination with the maximum stoichiometry could not be achieved due to limitations on the test bench. This meant that not all current-density points defined for the sensitivity test could be recorded for all conditions.

Temperature variations exhibit opposite effects on the local current density distribution in the investigated zones. With increasing temperature, the performance of zone 4 improves at all load points, showing a pronounced positive trend. This behavior can be attributed primarily to enhanced reaction kinetics. In contrast, zone 7 exhibits a more differentiated response. At low load points, an increase in temperature leads to a slight performance improvement. However, at higher load points, a slight performance decline is observed with increasing temperature. In zone 7, locally higher temperatures prevail, particularly at high loads, as a result of the increased temperature gradient between inlet and outlet. In combination with the drying effect of the anode inlet, this results in a sensitivity of the membrane resistance to the temperature, triggered by dehydration [44]. Consequently, a consistent behavior can be observed with the variation in humidification. Here, the performance in zone 7, which is close to the anode inlet, increases significantly with higher anode inlet gas humidity across the entire range. Again, this trend may be attributed to improved membrane hydration in this region, cf. e.g. [29].

In contrast, Zone 4 benefits only very slightly from increasing relative humidity. The fact that hardly any performance gain is observed in zone 4 with increasing relative humidity is likely attributable to the fact that the membrane in this region is already close to its saturation level at all investigated humidification conditions. Consequently, further increases in inlet gas humidity result in only minor additional membrane hydration

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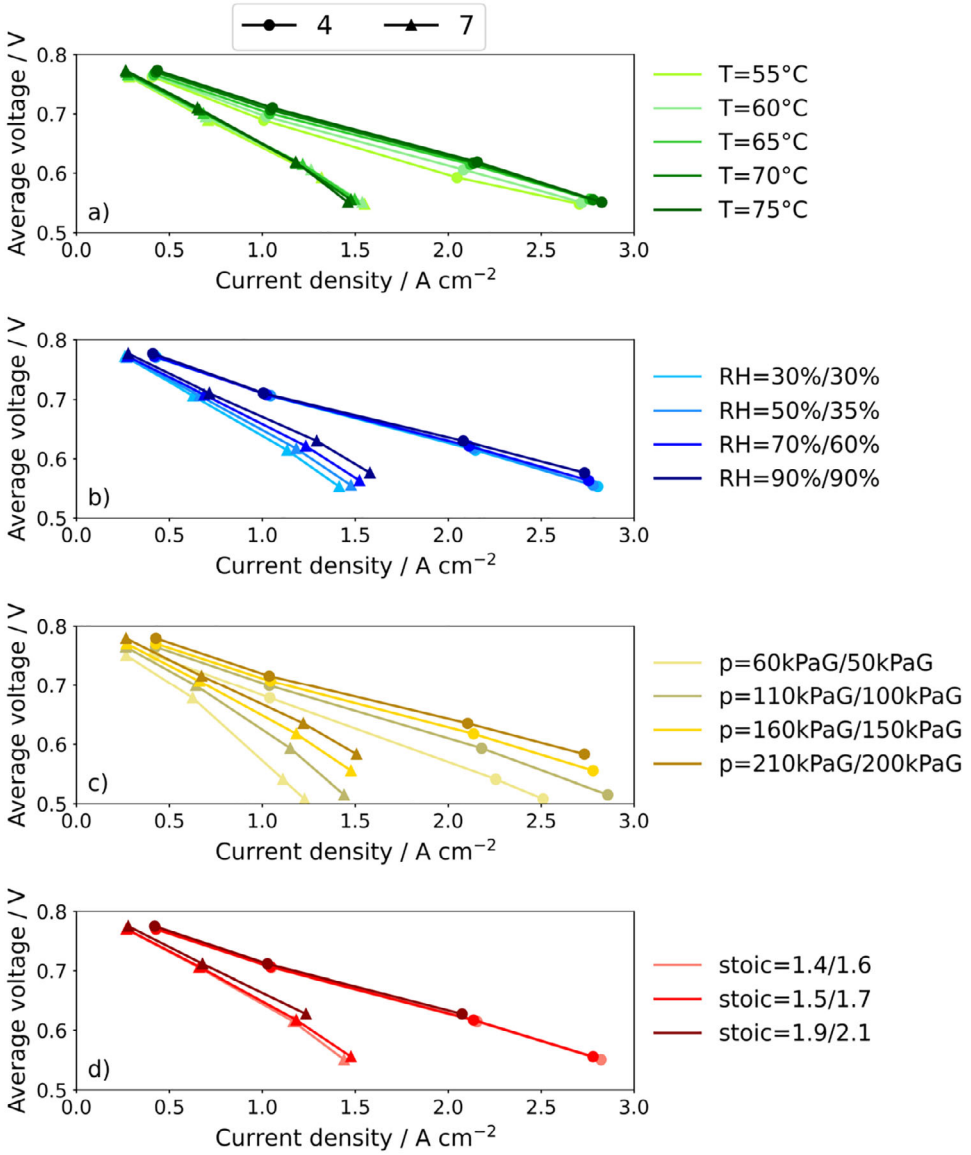


FIGURE 4 | Results of the sensitivity study for (from top to bottom) temperature, relative humidity, pressure, and stoichiometry for zone 4 (overall highest local current density) and zone 7 (lowest local current density).

and therefore provide only a very limited performance benefit. The variation in stoichiometry leads to rather minor and similar effects as the variation in humidification. The performance of zone 4 is hardly affected by the variation in stoichiometry, while the increase in stoichiometry has a positive effect on zone 7. This behavior may be attributed to the enhanced availability of oxygen in the cathode outlet regions at higher stoichiometries, which improves the local reactant supply in zone 7. In contrast, zone 4 appears to be sufficiently supplied with oxygen over the entire range of investigated stoichiometries, such that further increases do not lead to a significant performance enhancement [5, 6].

An interesting correlation can be seen in the variation in operating pressure. In contrast to the other operating conditions, pressure has a significant influence on both zones. An increase in operating pressure leads to higher local reactant concentrations in both zones, which, in turn, enhances the oxygen reduction reaction (ORR) kinetics [45]. As a result, a pronounced performance improvement is observed in both regions.

Figure 5 shows the differences in local current density between zone 4 and zone 7 over the different variations of the parameters' temperature, humidity, pressure, and stoichiometry at a load point of 1.8 A·cm^{-2}. The variations are listed according to the representation in Table 2. The respective values of the individual parameters are increasing in all cases with increasing variation identifier (consecutive no.).

It is clear that the different variations in operating conditions have different effects on heterogeneity. Temperature behaves in the opposite way to the other operating conditions; the heterogeneity increases with increasing temperature. In contrast, the current density distribution becomes more homogeneous with increasing humidity, increasing pressure, and increasing stoichiometry. The influence of humidity shown here supports the aforementioned

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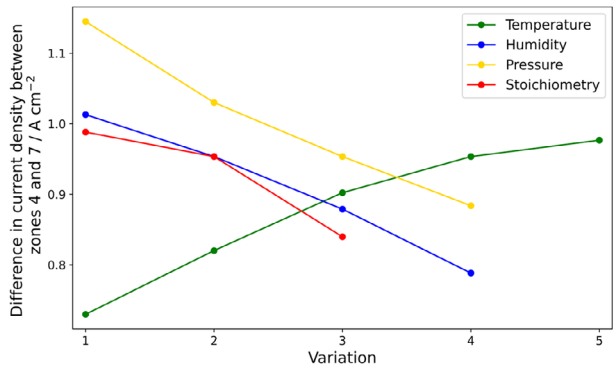


FIGURE 5 | Current density difference between zone 4 (highest current density) and zone 7 (lowest current density) at an average load of 1.8 A·cm^{-2} over the different variations of the sensitivity test (cf. Table 2) with increasing values over the variations in all cases.

interpretation of the different current density heat maps between descending and ascending polarization curves at low load points (see Figure 2). Overall, the operating pressure clearly has the greatest influence on the uniformity of the current density distribution.

The spatially resolved polarization curves clearly show differences between different zones and the ratio of minimum to maximum. However, this means that local behavior is lost, which can be depicted much better in 2D representations. These, in turn, offer only limited possibilities for comparing different operating parameter variations with each other. A suitable compromise is the representation along the direction of media flow. Figure 6 shows the averaged current density distribution along the cathode flow direction as part of the sensitivity analysis.

All four parameter variations examined show a similar basic pattern of current density distribution: At the cathode inlet, the current density is initially at a medium level, then rises in the middle of the cathode and drops significantly again toward the outlet. The temperature variation, in particular, shows that the most homogeneous distribution is achieved at 55°C. Surprisingly, however, the greatest inhomogeneity occurs at 70°C—an effect that was not apparent in the spatially resolved polarization curves. This makes clear that a higher temperature does not automatically lead to greater inhomogeneity. The other three parameters—humidity, pressure, and stoichiometry—confirm the findings described above: as humidity increases, pressure increases, and stoichiometry increases, the current density is distributed more evenly across the active area. The increasingly uniform current density distribution observed at higher humidity confirms that the shift in current density distribution at low load points, as described earlier in Figure 2, is primarily attributable to the membrane's moisture content. Looking at the representation along the cathode flow direction, parallels can

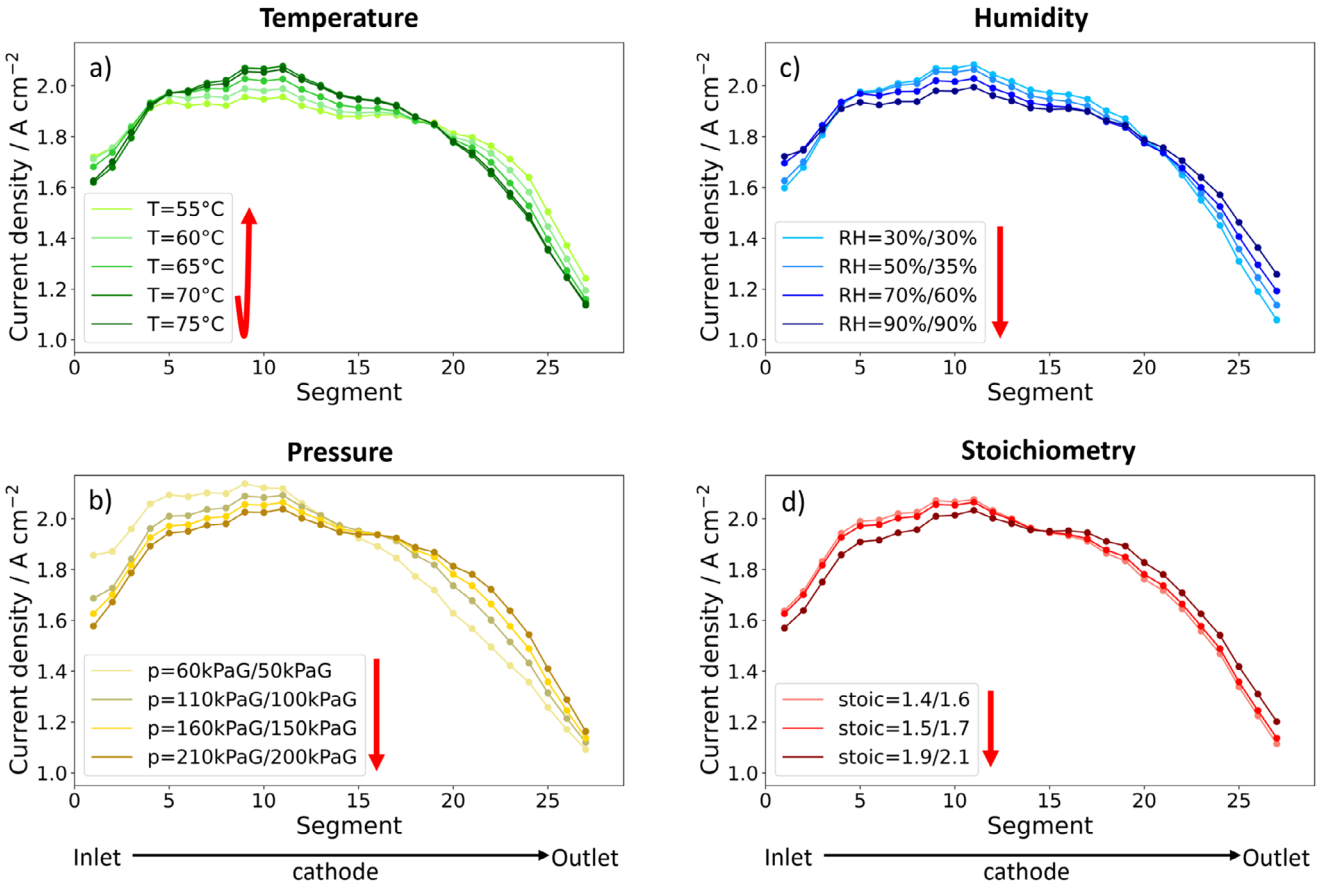


FIGURE 6 | Averaged current density distribution along the cathode flow direction of the temperature variation (a), pressure variation (b), humidity variation (c), and stoichiometry variation (d) at 1.8 A·cm^{-2}, with the increasing homogeneity of the current density distribution across the operating parameter variation indicated by a red arrow.

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also be seen between temperature and humidity on the one hand and between pressure and stoichiometry on the other. For humidity and temperature, there are two intersections in the curves of the parameters examined. The variants with the greatest inhomogeneity (T = 70°C and RH = 30%/30%) stand out in that they have the lowest current densities at the inlet and outlet, while reaching the highest values in the middle range. The decrease in current density in the cathode inlet area with decreasing humidity is due to the drying out of the membrane and the catalyst layer by the relatively dry inlet gas. This is in line with previous studies by Mitzel et al. [5]. In contrast to the temperature and humidity variation, there is only one intersection point for pressure and stoichiometry: in these cases, the lowest values at the inlet show the highest current densities, while the lowest values at the outlet show the lowest current densities. Increasing the operating pressure enhances the O_2 partial pressure and reduces the mass transport resistance. Thereby, the O_2 activity at the catalyst is enhanced, and the reaction rates toward the outlet are increased. Since the average current density remains constant, a current density increase in the outlet region will directly lead to a lowered current density in the remaining sections, resulting in a higher overall cell potential. Thus, the inlet region is no longer required to deliver this high share of the current density distribution as it is for low pressure. Consequently, effects that are minor for the lowest pressure, that is, the detailed shape of the humidity-determined membrane resistance and ORR resistance curves, come more into play for the inlet region. We will present a more detailed analysis of these phenomena based on locally resolved EIS data in a subsequent publication. For the stoichiometry variation, the current density at the cathode inlet also decreases with increasing stoichiometry. In the literature, it is discussed that the dominant mechanism is the increased air flow at higher stoichiometries [5, 6]. The higher air flow means that more oxygen is available for the reaction, which is particularly effective in the cathode outlet zone. A higher air flow also ensures that reaction water is transported out of the outlet area more effectively, thereby reducing flooding effects [6]. In contrast to the cathode inlet region, membrane moisture certainly plays a role. Increased flow of relatively dry air causes the membrane to dry out. The reduced local membrane hydration increases the ohmic resistance, causing the reaction zone to shift toward the cathode outlet.

3.2 | EIS

EIS measurements were performed to analyze the electrochemical processes and the distribution of relevant loss mechanisms. The results of the single-cell EIS are presented below, followed by a discussion of the spatially resolved EIS measurements for detailed investigation of local differences within the active area.

3.2.1 | Single Cell EIS

When performing EIS measurements at the stack level with large-format cells, various factors can limit data quality. In particular, the size and geometry of the stack and the arrangement of the cables can lead to capacitive or inductive effects in the high-frequency range. This is also the case in the experiments shown here (see Figure 7). To minimize the influence of such interference effects, the evaluated frequency range for the single-cell measurements was limited to 100 mHz–1026 Hz. The results of the EIS measurement at load points of 0.45, 0.9, and 1.8 A·cm^{-2} are shown in Figure 7a at cell 6, which is representative for all cells.

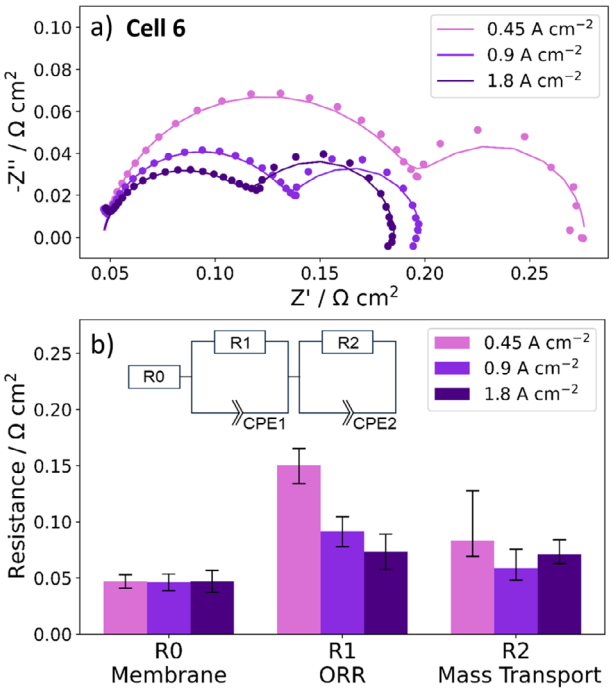


FIGURE 7 | Nyquist plots of single cell impedance spectra from cell 6 exemplarily at different current densities (a) and fitted resistances of the equivalent circuit fitting averaged over all 10 cells with marking of minimum and maximum values by error bar (b). R0: membrane resistance. R1: charge transfer resistance of the oxygen reduction reaction (ORR). R2: mass transport resistance.

The Nyquist plot shows two dominant processes. The high-frequency process is associated with the charge transfer of the ORR on the cathode side. In the lower frequency range, mass transport is limiting [23, 46–48]. The two dominant processes can be represented in an equivalent circuit as two R-CPE elements consisting of a resistor and a constant phase element (CPE). In addition to these processes, a significant shift of the curves due to ohmic losses can be seen in the Nyquist diagrams. The ohmic losses consist of an ionic and an electronic component. Heinzmann et al. [46] have shown that the ionic part, which results from proton transport in the MEA, is the dominant process. Accordingly, in addition to the two R-CPE elements, the equivalent circuit diagram includes another resistor that describes the membrane resistance. The complete equivalent circuit model and the averaged fit resistances across all cells, shown with error bars to indicate the minimum and maximum values, are shown in Figure 7b. Here, R0 describes the membrane resistance, R1 the charge transfer resistance of the ORR, and R3 the mass transport resistance.

The Nyquist plots already show that the total impedance decreases with increasing load, which is quantitatively confirmed by the fit parameters. The membrane resistance remains largely constant on average, but shows a slightly greater spread between

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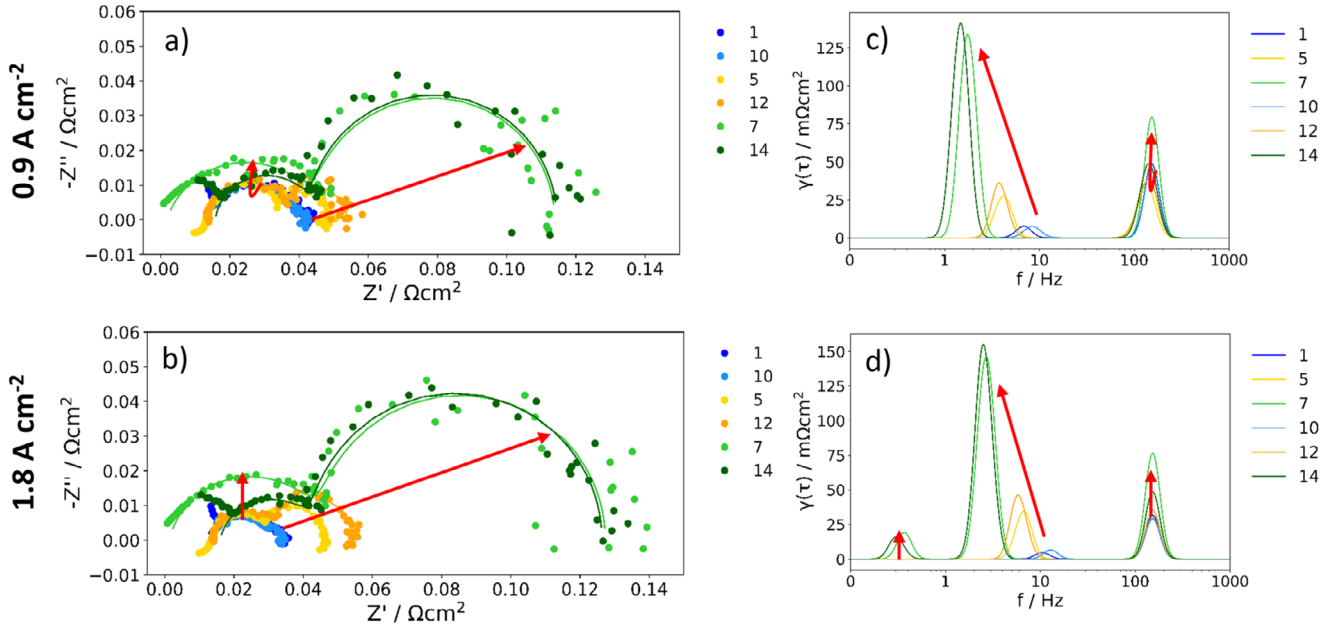


FIGURE 8 | Nyquist plots of spatially resolved EIS measurement of selected zones (a, b) and DRT analysis (c, d) at 0.9 A·cm^{-2} (a, c) and 1.8 A·cm^{-2} (b, d), with red arrows indicating the tendencies from cathode inlet toward cathode outlet.

the individual cells as the load increases. This suggests that differences in local wetting or temperature distribution become more pronounced at higher currents, since, as described in Section 3.1, the membrane resistance increases as the membrane dries out. The ORR resistance shows a characteristic curve as a function of current density. The ORR resistance decreases significantly with increasing load, which can be attributed to higher cathodic overpotentials that accelerate the electrochemical reaction [46, 48]. The mass transport resistance is lowest at 0.9 A cm^{-2} and increases at both lower and higher loads. At low loads, limited gas flow and water accumulation lead to higher apparent transport resistance, while at high loads, oxygen depletion and flooding effects in the cathode catalyst and gas diffusion layers hinder gas diffusion, resulting in a pronounced increase in mass transport resistance [48].

3.2.2 | Spatially Resolved EIS

The spatially resolved EIS measurements were performed simultaneously for each current density value in all 16 defined zones. Figure 8 shows the measured spectra for selected zones: cathode inlet (1 and 10), middle zones (5 and 12), and cathode outlet (7 and 16). The diagrams on the left show the selected spectra in the form of Nyquist plots, and the diagrams on the right show the evaluation of the spectra using the distribution of relaxation times (DRT) analysis method [46, 49]. A regulation parameter of $\lambda = 10^{-4}$ was used in the DRT analyses for all spectra examined. The parameter λ controls the balance between smoothness and resolution of the resulting DRT spectra. A smaller λ enhances resolution but increases sensitivity to noise, while a larger λ yields smoother results at the expense of detail. Therefore, λ must be carefully selected according to the data quality. For the DRT analysis and parameter fitting, the high-frequency range of the measurements was reduced due to capacitive and inductive artifacts. A maximum of 1180 Hz was set for inlet zones 1, 3, and 10. The upper frequency limit for the other zones was set to 517 Hz. The upper diagrams (Figure 8A,C) show the results for a load point of 0.9 A·cm^{-2}, while the lower diagrams (Figure 8B,D) correspond to 1.8 A·cm^{-2}. In all diagrams, red arrows for selected features illustrate the trend observed in the local zones from the cathode inlet to the cathode outlet.

The Nyquist diagram on the left shows two key processes at the cell level, which are also identified by the DRT analysis at the corresponding frequencies. The two processes can be assigned to the cell impedance of the ORR and the mass transport. The DRT analysis shows a significant increase in peak values from the cathode inlet to the cathode outlet, indicating that the impedances at the cathode outlet are higher than at the inlet. This observation is confirmed by the Nyquist plots: the cathode outlet areas marked in green show significantly higher impedance values than the middle areas marked in yellow and the inlet areas marked in blue. The shift of the frequency range of the mass transport process from the cathode inlet to the outlet toward lower frequencies in the DRT analysis shows that the gas transport conditions in the outlet area are significantly less favorable. The lower characteristic frequencies indicate slower transport dynamics and effectively longer diffusion paths, which lead to higher mass transport-related losses. A major advantage of DRT analysis over classic Nyquist plots is that the individual electrochemical processes can be clearly separated from each other, and no overlaps occur [46, 49]. This unambiguous discrimination reveals an additional third peak in the low-frequency range in the cathode outlet zones at a current density of 1.8 A·cm^{-2}. This can be attributed to the frequency range of diffusion losses. This result is consistent with the previously analyzed spatially resolved polarization curves, which also showed that deviations from ohmic-linear behavior already occur in the outlet zones under high load. The analysis confirms that limitations due to diffusion losses already become relevant at BoL in the outlet zones, while this is not yet observable in other areas.

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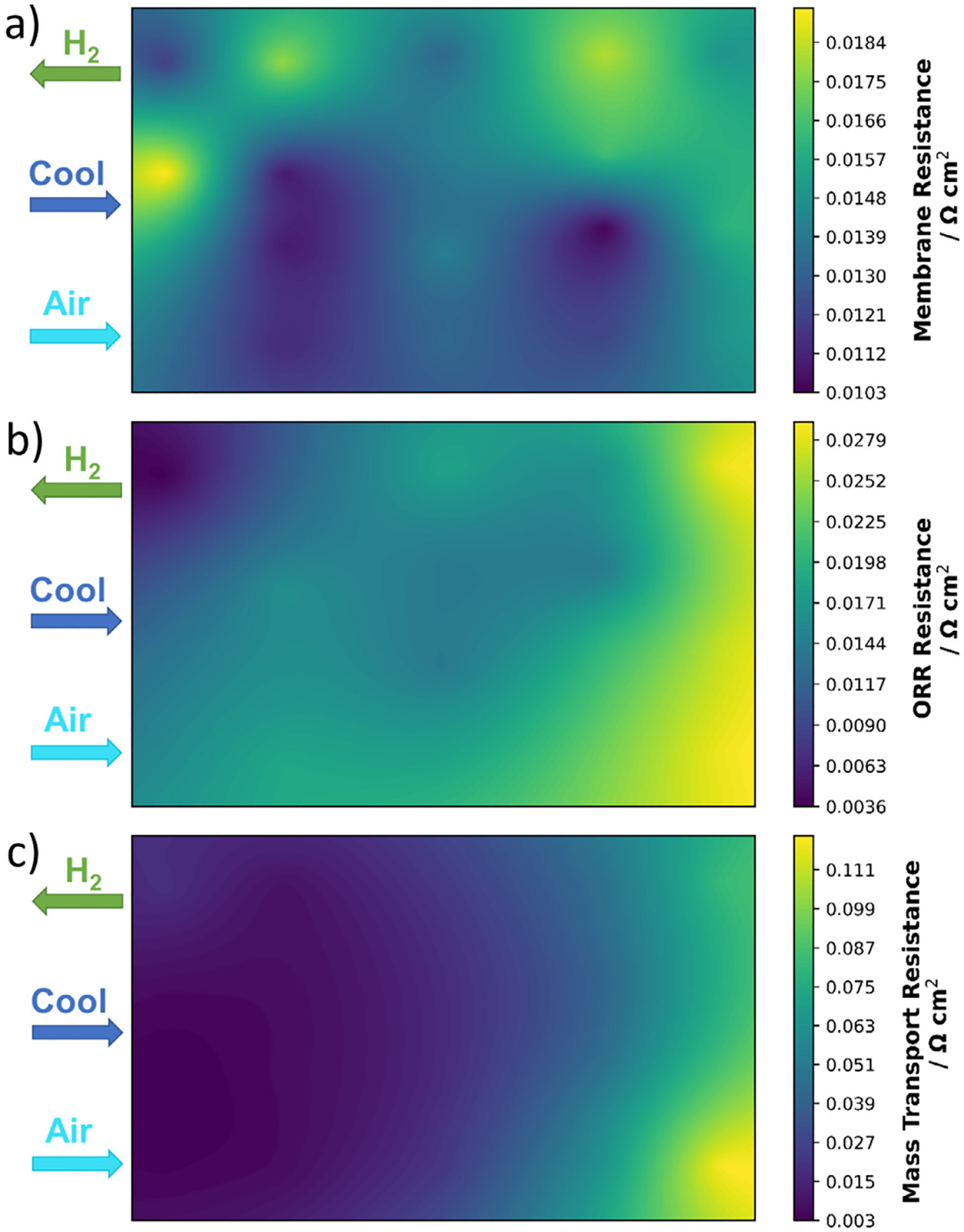


FIGURE 9 | Spatial distribution of resistances at 1.8 A·cm^{-2} fitted from spatially resolved impedance spectra across the active cell surface. The values were determined zone by zone, assigned to the respective zone centers, and linearly interpolated. To increase the resolution in the middle area, zones 5 and 12 were mirrored along the flow direction. (a) membrane resistance, (b) ORR resistance, (c) mass transport resistance.

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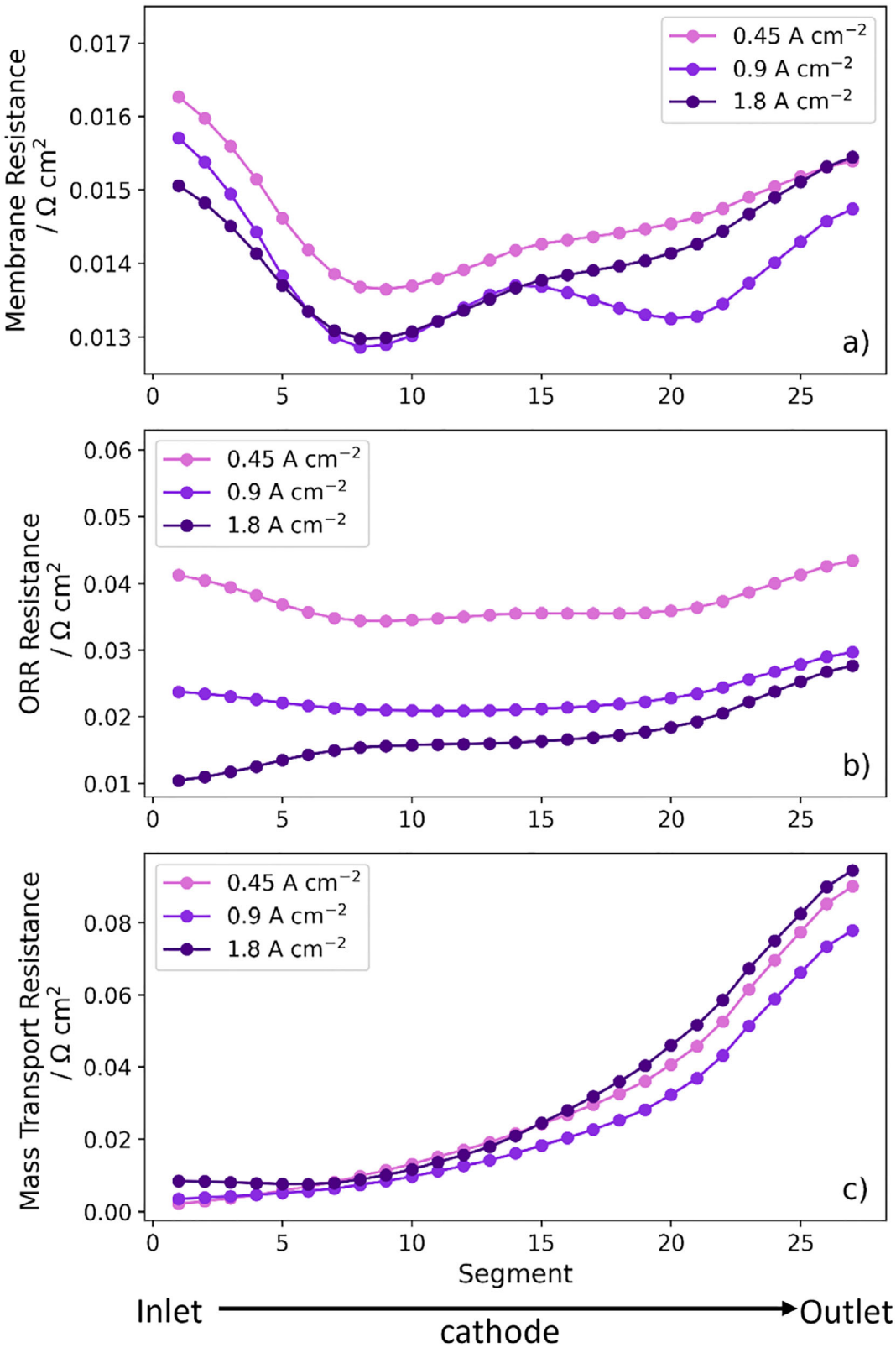


FIGURE 10 | Averaged membrane resistance (a), ORR resistance (b), mass transport resistance (c) over cathode flow field at 0.45, 0.9, and 1.8 A·cm^{-2}.

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The spatially resolved impedance spectra were fitted using the equivalent circuit diagram presented above, which consists of two R–CPE elements and an additional ohmic resistance. To visualize the resistance curves across the active area, the fitted values were first assigned to the centers of the respective zones and then linearly interpolated. For zone 4, the values from zone 9 were used because the measurements in zone 4 were faulty and did not provide realistic results. Similarly, the values from zone 16 were used for the membrane resistance and ORR resistance in zone 7, as inductive artifacts occurred in zone 7, which led to a distortion of these two resistance components. To achieve improved resolution in the middle of the active area, zones 5 and 12 were additionally mirrored along the flow direction. Since the current density distribution at these positions is comparable, it can be assumed that the impedance values are also equivalent. The resulting spatial distribution of the fitted resistance values at 1.8 $A{\cdot}cm^{-2}$ is shown in Figure 9.

The membrane resistance is lowest in the central area of the cell and increases toward the edges. The lower membrane resistance in the central area of the cell can be explained by the moisture balance of the membrane. Due to more uniform gas and water transport, the membrane remains better moistened in the center than at the edges [5–7, 9]. The ORR resistance increases continuously along the flow direction from the cathode inlet to the cathode outlet. The resistance at the edges is higher than in the central area. The increase from inlet to outlet correlates with oxygen depletion along the cathode. At the outlet, the oxygen concentration decreases, which limits the reaction kinetics locally [7, 8]. The mass transport resistance also increases significantly from the cathode inlet toward the cathode outlet. A local maximum is reached at the anode inlet. The increase in mass transport resistance is attributed to oxygen concentration gradients and water accumulation along the cathode flow field, which impede gas transport and limit reactant access to the catalyst sites [23, 47]. To compare the resistance values at different current densities, the interpolated 2D data over the active area was displayed as a 1D line plot. As with the current density distribution, a total of 27 areas were formed along the flow direction, and the mean value was calculated for each area. The resistance values across the flow direction for the three resistors are shown in Figure 10 for current densities of 0.45, 0.9, and 1.8 $A{\cdot}cm^{-2}$.

The membrane resistance shows a significant drop along the cathode flow at the cathode inlet and reaches its minimum at all load points in segments 8 and 9. The membrane resistance increases again toward the cathode outlet. At the inlet, the membrane resistance decreases with increasing current density, while at the outlet, it even reaches higher values at 1.8 $A{\cdot}cm^{-2}$ than at 0.45 $A{\cdot}cm^{-2}$. The largest spread between minimum and maximum membrane resistance was observed at 0.9 $A{\cdot}cm^{-2}$ with 0.00285 $\Omega\ cm^2$. The ORR resistance shows a significant decrease with increasing current density. This effect is particularly pronounced at the cathode inlet. At 1.8 $A{\cdot}cm^{-2}$, the ORR resistance increases continuously along the cathode from the inlet to the outlet. At the two lower current densities, however, a different behavior is observed: from the inlet to the middle areas, the resistance initially decreases and then increases again toward the outlet. At 0.9 $A{\cdot}cm^{-2}$, the drop at the inlet is already lower than at the lowest load point. With a maximum spread of 0.0173 $\Omega\ cm^2$ at 1.8 $A{\cdot}cm^{-2}$, the ORR resistance has a stronger influence on local differences than the membrane resistance. Even more pronounced is the variation in mass transport resistance, which shows its greatest spread of 0.088 $\Omega\ cm^2$ at 0.45 $A\ cm^{-2}$. Regardless of the current density, it shows a significant increase from the cathode inlet toward the cathode outlet. The lowest values over the entire channel length occur at 0.9 $A{\cdot}cm^{-2}$, while the highest values are reached at 1.8 $A\ cm^{-2}$.

4 | Conclusions

The results of this work contribute to a better understanding of local processes in fuel cells with large active areas, which are relevant for mobility applications in heavy-duty transport. Analysis of the current density distribution shows that significant inhomogeneities occur across the entire active area. In addition to the obvious dependence on the local oxygen concentration, the current density varies in particular with the moisture content of the membrane, which has a significant influence, especially at low current densities. At a load of 1.8 $A{\cdot}cm^{-2}$, local differences in current density of up to 0.938 $A{\cdot}cm^{-2}$ were observed. Among the investigated operating parameters, temperature, humidity, pressure, and stoichiometry, pressure proved to be the dominant factor influencing the homogeneity of the current density distribution.

Furthermore, this work highlights the added value of spatially resolved EIS analysis, which provides new insights into the local electrochemical processes within a stack with an active area of 280 cm^2. The EIS measurements confirm the existence of significant local differences across the entire active area. A clear trend can be seen in the increase in impedance values from the cathode inlet to the cathode outlet. The membrane resistance is strongly influenced by the moisture balance, reaching its lowest values in the middle of the cell and increasing toward the edges. The ORR resistance decreases significantly with increasing load, both at the cell level and across the active area, with this effect being particularly pronounced at the cathode inlet. The mass transport resistance shows the largest increase from the cathode inlet to the cathode outlet, which is due to increasingly limited reaction kinetics along the cathode gas flow.

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References

1. *Greenhouse Gas Emissions From Transport in Europe*, https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-transport?activeAccordion=, 2024.

2. *Reducing Greenhouse Gas Emissions From Heavy-Duty Vehicles in Europe*, https://www.eea.europa.eu/en/analysis/publications/reducing-greenhouse-gas-emissions-from-heavy-duty-vehicles-in-europe, 2022.

16156854, 2026, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/fuce.70057 by Eva Fensterle - Otto-Von-Taube-Gymn , Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3. Y. Ruf, M. Baum, T. Zorn, A. Menzel, and J. Rehberger *Fuel Cells Hydrogen Trucks Heavy-Duty's High Performance Green Solution* (2020), https://www.clean-hydrogen.europa.eu/document/download/81edb800-d1fc-49f4-8341-c894d3ab77e4_en?filename=FCH%20HDT%20-%20Study%20Report_final_vs.pdf.

4. D. A. Cullen, K. C. Neyerlin, and R. K. Ahluwalia, et al., "New Roads and Challenges for Fuel Cells in Heavy-Duty Transportation," *National Energy* 6 (2021): 462–474, https://doi.org/10.1038/s41560-021-00775-z.

5. J. Mitzel, et al., "Fault Diagnostics in PEMFC Stacks by Evaluation of Local Performance and Cell Impedance Analysis," *Fuel Cells* (2020): 403–412, https://doi.org/10.1002/fuce.201900193.

6. Q. Zhang, R. Lin, L. Técher, and X. Cui, "Experimental Study of Variable Operating Parameters Effects on Overall PEMFC Performance and Spatial Performance Distribution," *Energy* 115 (2016): 550–560, https://doi.org/10.1016/j.energy.2016.08.086.

7. E. Engebretsen, G. Hinds, and Q. Meyer, et al., "Localised Electrochemical Impedance Measurements of a Polymer Electrolyte Fuel Cell Using a Reference Electrode Array to Give Cathode-Specific Measurements and Examine Membrane Hydration Dynamics," *Journal of Power Sources* 382 (2018): 38–44, https://doi.org/10.1016/j.jpowsour.2018.02.022.

8. S. Shimpalee, S. Hirano, M. DeBolt, V. Lilavivat, J. W. Weidner, and Y. Khunatorn, "Macro-Scale Analysis of Large Scale PEM Fuel Cell Flow-Fields for Automotive Applications," *Journal of the Electrochemical Society* 164 (2017): E3073, https://doi.org/10.1149/2.0091711jes.

9. I. S. Hussaini and C.-Y. Wang, "Transients of Water Distribution and Transport in PEM Fuel Cells," *Journal of Electrochemical Society* 156 (2009): B1394, https://doi.org/10.1149/1.3236646.

10. Q. Meyer, et al., "Combined Current and Temperature Mapping in an Air-Cooled, Open-Cathode Polymer Electrolyte Fuel Cell Under Steady-State and Dynamic Conditions," *Journal of Power Sources* 297 (2015): 315–322, https://doi.org/10.1016/j.jpowsour.2015.07.069.

11. L. Xing, Y. Wang, P. K. Das, K. Scott, and W. Shi, "Homogenization of Current Density of PEM Fuel Cells by In-Plane Graded Distributions of Platinum Loading and GDL Porosity," *Chemical Engineering Science* 192 (2018): 699–713, https://doi.org/10.1016/j.ces.2018.08.029.

12. J. Sim, M. Kang, and K. Min, "Effects of Porosity Gradient and Average Pore Size in the In-Plane Direction and Disposition of Perforations in the Gas Diffusion Layer on the Performance of Proton Exchange Membrane Fuel Cells," *Journal of Power Sources* 544 (2022): 231912, https://doi.org/10.1016/j.jpowsour.2022.231912.

13. Y. Zhang, A. Smirnova, A. Verma, and R. Pitchumani, "Design of a Proton Exchange Membrane (PEM) Fuel Cell with Variable Catalyst Loading," *Journal of Power Sources* 291 (2015): 46–57, https://doi.org/10.1016/j.jpowsour.2015.05.002.

14. A. A. Kulikovsky, "Optimal Shape of Catalyst Loading Along the Oxygen Channel of a PEM Fuel Cell," *Electrochimica Acta* 54 (2009): 7001–7005, https://doi.org/10.1016/j.electacta.2009.07.005.

15. Y. Wang, X. Wang, Y. Fan, W. He, J. Guan, and X. Wang, "Numerical Investigation of Tapered Flow Field Configurations for Enhanced Polymer Electrolyte Membrane Fuel Cell Performance," *Applied Energy* 306 (2022): 118021.

16. X. Zhang, X. Yang, W. Gao, and C. Wang, "An Experimental Research on the Net Output Power and Current Density Distribution of PEM Fuel Cells With Trapezoid Baffled Flow Fields," *International Journal of Energy Research* 45 (2021): 21464–21475, https://doi.org/10.1002/er.7194.

17. C. Yin, J. Gao, and X. Wen, et al., "In Situ Investigation of Proton Exchange Membrane Fuel Cell Performance With Novel Segmented Cell Design and a Two-Phase Flow Model," *Energy* 113 (2016): 1071–1089, https://doi.org/10.1016/j.energy.2016.06.097.

18. C. Yin, Y. Gao, K. Li, Y. Song, and H. Tang, "Experimental Investigation on Local Behaviors of PEMFC with Segmented Cell," *Automotive Innovation* 4 (2021): 165–175.

19. W. Tang, R. Lin, Y. Weng, J. Zhang, and J. Ma, "The Effects of Operating Temperature on Current Density Distribution and Impedance Spectroscopy by Segmented Fuel Cell," *International Journal of Hydrogen Energy* 38 (2013): 10985–10991, https://doi.org/10.1016/j.ijhydene.2013.01.099.

20. Z. Liu, Z. Mao, B. Wu, L. Wang, and V. M. Schmidt, "Current Density Distribution in PEFC," *Journal of Power Sources* 141 (2005): 205–210, https://doi.org/10.1016/j.jpowsour.2004.10.003.

21. M. Schüttoff, C. Wachtel, R. Schlumberger, F. Wilhelm, J. Scholta, and M. Hölzle, "Development of Accelerated Durability Test Protocols for Polymer Electrolyte Membrane Fuel Cell Stacks Under Realistic Operating Conditions," *Fuel Cells* 24: (2024): e202300263.

22. X. Zhang, Y. Yang, L. Guo, and H. Liu, "Performance Degradation of Proton Exchange Membrane Fuel Cell Caused by an Accelerated Stress Test," *Fuel Cells* 19 (2019): 160–168, https://doi.org/10.1002/fuce.201800152.

23. G. Ustuner, Y. Hung, and D. Mahajan, "Investigation of Proton Exchange Membrane Fuel Cell Performance by Exploring the Synergistic Effects of Reaction Parameters via Power Curve and Impedance Spectroscopy Analysis," *Energies* 17 (2024): 2530.

24. X. Li, Z. Ge, and R. Zhang, et al., "Measurement and Analysis on Current Density Distribution of Long-Channel Proton Exchange Membrane Fuel Cell: Insights From Operational Conditions and Electrochemical Impedance," *Energy* 318 (2025): 134790, https://doi.org/10.1016/j.energy.2025.134790.

25. N. Wagner, "Characterization of Membrane Electrode Assemblies in Polymer Electrolyte Fuel Cells Using a.c. Impedance Spectroscopy," *Journal of Applied Electrochemistry* (2002): 859–863, https://doi.org/10.1023/A:1020551609230.

26. I. A. Schneider, H. Kuhn, A. Wokaun, and G. G. Scherer, "Study of Water Balance in a Polymer Electrolyte Fuel Cell by Locally Resolved Impedance Spectroscopy," *Journal of The Electrochemical Society* 152 (2005): A2383–A2389, https://doi.org/10.1149/1.2121739.

27. I. A. Schneider, M. H. Bayer, and S. von Dahlen, "Locally Resolved Electrochemical Impedance Spectroscopy in Channel and Land Areas of a Differential Polymer Electrolyte Fuel Cell," *Journal of The Electrochemical Society* 158 (2011): B343–B348, https://doi.org/10.1149/1.3536498.

28. I. A. Schneider, H. Kuhn, A. Wokaun, and G. G. Scherer, "Fast Locally Resolved Electrochemical Impedance Spectroscopy in Polymer Electrolyte Fuel Cells," *Journal of The Electrochemical Society* 152 (2005): A2092–A2103, https://doi.org/10.1149/1.2032409.

29. D. Gerteisen, W. Mérida, T. Kurz, P. Lupotto, M. Schwager, and C. Hebling, "Spatially Resolved Voltage, Current and Electrochemical Impedance Spectroscopy Measurements," *Fuel Cells* 11 (2011): 339–349, https://doi.org/10.1002/fuce.201000181.

30. D. Gerteisen, N. Zamel, C. Sadeler, F. Geiger, V. Ludwig, and C. Hebling, "Effect of Operating Conditions on Current Density Distribution and High Frequency Resistance in a Segmented PEM Fuel Cell," *International Journal of Hydrogen Energy* 37 (2012): 7736–7744, https://doi.org/10.1016/j.ijhydene.2012.02.024.

31. D. Liu, R. Lin, and B. Feng, et al., "Localised Electrochemical Impedance Spectroscopy Investigation of Polymer Electrolyte Membrane Fuel Cells Using Print Circuit Board Based Interference-Free System," *Applied Energy* 254 (2019): 113712, https://doi.org/10.1016/j.apenergy.2019.113712.

32. D. Liu, R. Lin, B. Feng, and Z. Yang, "Investigation of the Effect of Cathode Stoichiometry of Proton Exchange Membrane Fuel Cell Using Localized Electrochemical Impedance Spectroscopy Based on Print Circuit Board," *International Journal of Hydrogen Energy* 44 (2019): 7564–7573, https://doi.org/10.1016/j.ijhydene.2019.01.095.

33. D. J. L. Brett, S. Atkins, N. P. Brandon, V. Vesovic, N. Vasileiadis, and A. Kucernak, "Localized Impedance Measurements along a Single Channel of a Solid Polymer Fuel Cell," *The Electrochemical Society* 6 (2003): A63–A66.

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34. F. Haimerl, J. P. Sabawa, T. A. Dao, and A. S. Bandarenka, "Spatially Resolved Electrochemical Impedance Spectroscopy of Automotive PEM Fuel Cells," *ChemElectroChem* 9 (2022): e202200069.

35. F. Häußler, B. Wiedemann, C. Bergbreiter, et al., *The ZSW HyFaB Generic Stack*, 2024, Zentrum fuer Sonnenenergie- und Wasserstoff-Forschung Baden-Wuerttemberg (ZSW), https://zenodo.org/records/14223364/files/2024-11-29_Presentation_HyFaB_and_GenStack.pdf.

36. Q. Zhang, M. Schulze, P. Gazdzicki, and K. A. Friedrich, "Comparison of Different Performance Recovery Procedures for Polymer Electrolyte Membrane Fuel Cells," *Applied Energy* 302 (2021): 117490.

37. X.-Z. Yuan, C. Song, H. Wang, and J. Zhang *Electrochemical Impedance Spectroscopy in PEM Fuel Cells. Fundamentals and Application* (Springer, 2009).

38. S. Schneider, F. Wilhelm, J. Scholta, M. Schüttoff, and L. Jörissen, "Further Development of an OpenFOAM LT-PEMFC Toolbox and Its Validation on an Automotive Fuel Cell Design," *Energies* 18 (2025): 4793.

39. J. Mitzel, E. Gülzow, and A. Kabza, et al., "Identification of Critical Parameters for PEMFC Stack Performance Characterization and Control Strategies for Reliable and Comparable Stack Benchmarking," *International Journal of Hydrogen Energy* 41 (2016): 21415–21426, https://doi.org/10.1016/j.ijhydene.2016.08.065.

40. M. Koprek, R. Schlumberger, and C. Wachtel, et al., "Local Ageing Effects of Polymer Electrolyte Fuel Cell Membrane Electrode Assemblies due to Accelerated Durability Testing," *Fuel Cells* 20 (2022): 271–283, https://doi.org/10.1002/fuce.202200064.

41. A New Load Cycle and Durability Test Procedure Based on Vehicle Fleet Data: ID-Fast Final Workshop, https://www.id-fast.eu/uploads/media/ID-FAST_Final_Workshop_2021-12-16_03_Drive_Cycle_public.pdf, 2021.

42. R. Zhang, B. Yang, and X. Lei, et al., "Droplets Dynamics Theory and Micro-Flow Field Experiments of Improving Self-Humidifying Feature and Maximum Power Density in Fuel Cells," *Chemical Engineering Journal* 429 (2022): 131974, https://doi.org/10.1016/j.cej.2021.131974.

43. EG&G Technical Services, Inc., Fuel Cell Handbook (2004), https://www.netl.doe.gov/sites/default/files/netl-file/FCHandbook7.pdf.

44. P. Qi, Z. Wu, and J. Mou, et al., "A Review of Water Management in Proton Exchange Membrane Fuel Cell Systems," *Sustainable Energy Fuels* 9 (2024): 72–97, https://doi.org/10.1039/D4SE01020E.

45. L. Wang, A. Husar, T. Zhou, and H. Liu, "A Parametric Study of PEM Fuel Cell Performances," *International Journal of Hydrogen Energy* 28 (2003): 1263–1272, https://doi.org/10.1016/S0360-3199(02)00284-7.

46. M. Heinzmann, A. Weber, and E. Ivers-Tiffée, "Advanced Impedance Study of Polymer Electrolyte Membrane Single Cells by Means of Distribution of Relaxation Times," *Journal of Power Sources* 402 (2018): 24–33, https://doi.org/10.1016/j.jpowsour.2018.09.004.

47. L. Zhao, H. Dai, F. Pei, P. Ming, X. Wei, and J. Zhou, "A Comparative Study of Equivalent Circuit Models for Electro-Chemical Impedance Spectroscopy Analysis of Proton Exchange Membrane Fuel Cells," *Energies* (2022): 386, https://doi.org/10.3390/en15010386.

48. J. Santana, M. Espinoza-Andaluz, T. Li, and M. Andersson, "A Detailed Analysis of Internal Resistance of a PEFC Comparing High and Low Humidification of the Reactant Gases," *Frontiers in Energy Research* 8 (2020), https://doi.org/10.3389/fenrg.2020.00217.

49. S. Dierickx, A. Weber, and E. Ivers-Tiffée, "How the Distribution of Relaxation Times Enhances Complex Equivalent Circuit Models for Fuel Cells," *Electrochimica Acta* 355 (2020): 136764, https://doi.org/10.1016/j.electacta.2020.136764.

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